PRN: 260840820008

Name: Arnav Ajay Umarkar

Section 1) MySQL Queries

Question 1)

```
KD4_98781_Arnav>SELECT
    ->     book_name
    ->     , publish_date
    -> FROM
    ->     books
    -> WHERE
    ->     YEAR(publish_date) > 2000
    -> ORDER BY
    ->     publish_date ASC;
+--------------------+--------------+
| book_name          | publish_date |
+--------------------+--------------+
| Wise and Otherwise | 2002-03-15   |
| Da Vinci Code      | 2003-04-01   |
| Five Point Someone | 2004-05-15   |
| Immortals of Meluha | 2010-02-01   |
| Half Girlfriend    | 2014-10-01   |
+--------------------+--------------+
5 rows in set (0.03 sec)
```

Question 2)

```
KD4_98781_Arnav>SELECT
    ->     UPPER(book_name)
    ->     , LENGTH(REPLACE(book_name, " ", "")) AS length_without_spaces
    -> FROM
    ->     books;
+--------------------+----------------------+
| UPPER(book_name)   | length_without_spaces |
+--------------------+----------------------+
| MALGUDI DAYS       |                   11 |
| FIVE POINT SOMEONE |                   16 |
| HARRY POTTER       |                   11 |
| 1984               |                    4 |
| IMMORTALS OF MELUHA |                   17 |
| WISE AND OTHERWISE |                   16 |
| DA VINCI CODE      |                   11 |
| THE BLUE UMBRELLA  |                   15 |
| HALF GIRLFRIEND    |                   14 |
| HARRY POTTER 2     |                   12 |
+--------------------+----------------------+
10 rows in set (0.03 sec)
```

Question 3)

```
KD4_98781_Arnav>SELECT
    ->     book_name
    ->     , price
    -> FROM
    ->     books
    -> ORDER BY
    ->     price DESC;
+---------------------+--------+
| book_name           | price  |
+---------------------+--------+
| Harry Potter 2      | 700.00 |
| Da Vinci Code       | 650.00 |
| Harry Potter        | 600.00 |
| Immortals of Meluha | 550.00 |
| 1984                | 500.00 |
| Malgudi Days        | 450.00 |
| Wise and Otherwise  | 400.00 |
| Half Girlfriend     | 380.00 |
| Five Point Someone  | 350.00 |
| The Blue Umbrella   | 300.00 |
+---------------------+--------+
10 rows in set (0.00 sec)
```

Question 4)

```
KD4_98781_Arnav>SELECT
    ->     category
    ->     , COUNT(book_id)
    -> FROM
    ->     books
    -> GROUP BY
    ->     category
    -> ORDER BY
    ->     COUNT(book_id) DESC;
+-----------+----------------+
| category  | COUNT(book_id) |
+-----------+----------------+
| Fiction   |              4 |
| Fantasy   |              2 |
| Novel     |              1 |
| Mythology |              1 |
| Thriller  |              1 |
| Romance   |              1 |
+-----------+----------------+
6 rows in set (0.04 sec)
```

Question 5)

```
KD4_98781_Arnav>SELECT
    ->     category
    ->     , COUNT(book_id) AS count
    ->     , AVG(price) AS avgPrice
    ->     , MAX(price) AS maxPrice
    -> FROM
    ->     books
    -> GROUP BY
    ->     category
    -> ORDER BY
    ->     AVG(price) DESC;
+-----------+-------+------------+----------+
| category  | count | avgPrice   | maxPrice |
+-----------+-------+------------+----------+
| Fantasy   |     2 | 650.000000 |   700.00 |
| Thriller  |     1 | 650.000000 |   650.00 |
| Mythology |     1 | 550.000000 |   550.00 |
| Novel     |     1 | 500.000000 |   500.00 |
| Romance   |     1 | 380.000000 |   380.00 |
| Fiction   |     4 | 375.000000 |   450.00 |
+-----------+-------+------------+----------+
6 rows in set (0.04 sec)
```

Question 6)

```
KD4_98781_Arnav>SELECT
    ->     category
    ->     , COUNT(book_id)
    -> FROM
    ->     books
    -> GROUP BY
    ->     category
    -> HAVING
    ->     COUNT(book_id) >= 2
    -> ORDER BY
    ->     COUNT(book_id) DESC;
+----------+---------------+
| category | COUNT(book_id) |
+----------+---------------+
| Fiction  |             4 |
| Fantasy  |             2 |
+----------+---------------+
2 rows in set (0.00 sec)
```

Question 7)

```
KD4_98781_Arnav>SELECT
    ->     author_name
    ->     , COUNT(b.book_id) AS countOfBooks
    -> FROM
    ->     authors AS a
    -> LEFT JOIN
    ->     books AS b
    -> ON
    ->     a.author_id = b.author_id
    -> GROUP BY
    ->     a.author_id, a.author_name;
+----------------+--------------+
| author_name    | countOfBooks |
+----------------+--------------+
| R K Narayan    |            1 |
| Chetan Bhagat  |            2 |
| J K Rowling    |            2 |
| George Orwell  |            1 |
| Amish Tripathi |            1 |
| Sudha Murty    |            1 |
| Dan Brown      |            1 |
| Ruskin Bond    |            1 |
| Arundhati Roy  |            0 |
+----------------+--------------+
9 rows in set (0.00 sec)
```

Question 8)

```
KD4_98781_Arnav>SELECT
    ->     *
    -> FROM
    ->     books AS b
    -> WHERE
    ->     b.price > (SELECT AVG(price) FROM books);
+---------+--------------------+-----------+--------+--------------+-----------+----------+
| book_id | book_name          | category  | price  | publish_date | author_id | quantity |
+---------+--------------------+-----------+--------+--------------+-----------+----------+
|     203 | Harry Potter       | Fantasy   | 600.00 | 1997-06-26   |       103 |       10 |
|     204 | 1984               | Novel     | 500.00 | 1949-06-08   |       104 |        4 |
|     205 | Immortals of Meluha | Mythology | 550.00 | 2010-02-01   |       105 |        7 |
|     207 | Da Vinci Code      | Thriller  | 650.00 | 2003-04-01   |       107 |        5 |
|     210 | Harry Potter 2     | Fantasy   | 700.00 | 1998-07-02   |       103 |        8 |
+---------+--------------------+-----------+--------+--------------+-----------+----------+
5 rows in set (0.00 sec)
```

Question 9)

```
KD4_98781_Arnav>SELECT
    ->     *
    -> FROM
    ->     books AS b
    -> WHERE
    ->     b.price < (SELECT MAX(price) FROM books)
    -> ORDER BY
    ->     b.price DESC;
+---------+--------------------+-----------+--------+--------------+-----------+----------+
| book_id | book_name          | category  | price  | publish_date | author_id | quantity |
+---------+--------------------+-----------+--------+--------------+-----------+----------+
|     207 | Da Vinci Code      | Thriller  | 650.00 | 2003-04-01   |       107 |        5 |
|     203 | Harry Potter       | Fantasy   | 600.00 | 1997-06-26   |       103 |       10 |
|     205 | Immortals of Meluha | Mythology | 550.00 | 2010-02-01   |       105 |        7 |
|     204 | 1984               | Novel     | 500.00 | 1949-06-08   |       104 |        4 |
|     201 | Malgudi Days       | Fiction   | 450.00 | 1982-01-10   |       101 |        5 |
|     206 | Wise and Otherwise | Fiction   | 400.00 | 2002-03-15   |       106 |        6 |
|     209 | Half Girlfriend    | Romance   | 380.00 | 2014-10-01   |       102 |        6 |
|     202 | Five Point Someone | Fiction   | 350.00 | 2004-05-15   |       102 |        8 |
|     208 | The Blue Umbrella  | Fiction   | 300.00 | 1974-01-20   |       108 |        9 |
+---------+--------------------+-----------+--------+--------------+-----------+----------+
9 rows in set (0.00 sec)
```

Question 10)

```
KD4_98781_Arnav>SELECT
    ->     m.member_name
    ->     , bks.book_name
    ->     , b.borrow_date
    ->     , b.status
    -> FROM
    ->     borrowings AS b
    -> INNER JOIN
    ->     members AS m
    -> ON
    ->     b.member_id = m.member_id
    -> INNER JOIN
    ->     books AS bks
    -> ON
    ->     b.book_id = bks.book_id
    -> INNER JOIN
    ->     authors AS a
    -> ON
    ->     a.author_id = bks.author_id
    -> WHERE
    ->     b.return_date IS NULL;
+-------------+--------------------+-------------+----------+
| member_name | book_name          | borrow_date | status   |
+-------------+--------------------+-------------+----------+
| Meena       | Harry Potter 2     | 2025-03-05  | Borrowed |
| Kiran       | Immortals of Meluha | 2025-02-05  | Borrowed |
| Raj         | Immortals of Meluha | 2025-03-15  | Borrowed |
| Raj         | Da Vinci Code      | 2025-02-15  | Borrowed |
+-------------+--------------------+-------------+----------+
4 rows in set (0.00 sec)
```

Question 11)

```
KD4_98781_Arnav>        SELECT
    ->     a.author_name
    ->     , AVG(price)
    -> FROM
    ->     authors AS a
    -> INNER JOIN
    ->     books AS b
    -> ON
    ->     a.author_id = b.author_id
    -> GROUP BY
    ->     b.author_id, a.author_name
    -> HAVING
    ->     AVG(price) > 500
    -> ORDER BY
    ->     AVG(price) DESC;
+---------------+------------+
| author_name    | AVG(price) |
+---------------+------------+
| J K Rowling    | 650.000000 |
| Dan Brown      | 650.000000 |
| Amish Tripathi | 550.000000 |
+---------------+------------+
3 rows in set (0.00 sec)
```

Question 12)

```
KD4_98781_Arnav>SELECT
    ->     a.author_name
    ->     , COUNT(b.book_id)
    ->     , AVG(price)
    -> FROM
    ->     authors AS a
    -> INNER JOIN
    ->     books AS b
    -> ON
    ->     a.author_id = b.author_id
    -> GROUP BY
    ->     b.author_id, a.author_name
    -> HAVING
    ->     AVG(price) > (SELECT AVG(b1.price) FROM books AS b1)
    -> ORDER BY
    ->     AVG(price) DESC;
+---------------+-----------------+------------+
| author_name    | COUNT(b.book_id) | AVG(price) |
+---------------+-----------------+------------+
| J K Rowling    |               2 | 650.000000 |
| Dan Brown      |               1 | 650.000000 |
| Amish Tripathi |               1 | 550.000000 |
| George Orwell  |               1 | 500.000000 |
+---------------+-----------------+------------+
4 rows in set (0.03 sec)
```

Section 2) Stored Procedure

Question 1)



s

SECTION 3) CTE Window function

1)

```
KD4_98781_Arnav>WITH memberships AS (
    -> SELECT
    ->   Membership_type
    -> , COUNT(member_id)
    -> , avg(age)
    -> FROM
    ->      Members as m
    -> GROUP BY
    ->      m.membership_type
    -> HAVING
    ->  COUNT(member_id) > 2
    -> And AVG(AGE) > 25
    -> )
    -> SELECT * FROM memberships;
+-----------------+-------------------+----------+
| Membership_type | COUNT(member_id) | avg(age) |
+-----------------+-------------------+----------+
| Gold            |                 5 |  30.0000 |
+-----------------+-------------------+----------+
1 row in set (0.04 sec)
```

2)

```
KD4_98781_Arnav>SELECT
    ->  Book_name
    ->  , category
    ->  ,price
    ->  , ROW_NUMBER() OVER(PARTITION BY category order by price)
    ->  , dense_rank() over(partition by category order by price)
    -> FROM
    ->  Books;
+-----------------------+-----------+--------+-----------------------------------------------------+-----------------------------------------------------+
| Book_name             | category  | price  | ROW_NUMBER() OVER(PARTITION BY category order by price) | dense_rank() over(partition by category order by price) |
+-----------------------+-----------+--------+-----------------------------------------------------+-----------------------------------------------------+
| Harry Potter          | Fantasy   | 600.00 |                                                   1 |                                                   1 |
| Harry Potter 2        | Fantasy   | 700.00 |                                                   2 |                                                   2 |
| The Blue Umbrella     | Fiction   | 300.00 |                                                   1 |                                                   1 |
| Five Point Someone    | Fiction   | 350.00 |                                                   2 |                                                   2 |
| Wise and Otherwise    | Fiction   | 400.00 |                                                   3 |                                                   3 |
| Malgudi Days          | Fiction   | 450.00 |                                                   4 |                                                   4 |
| Immortals of Meluha   | Mythology | 550.00 |                                                   1 |                                                   1 |
| 1984                  | Novel     | 500.00 |                                                   1 |                                                   1 |
| Half Girlfriend       | Romance   | 380.00 |                                                   1 |                                                   1 |
| Da Vinci Code         | Thriller  | 650.00 |                                                   1 |                                                   1 |
+-----------------------+-----------+--------+-----------------------------------------------------+-----------------------------------------------------+
10 rows in set (0.03 sec)

KD4_98781_Arnav>
```

3)

```
KD4_98781_Arnav>SELECT
    ->  b.book_id
    ->  , bks.book_name
    ->  , count(b.book_id)
    -> FROM
    ->  Borrowings as b
    -> Inner join
    ->  books as bks
    -> On
    ->  b.book_id = bks.book_id
    -> group by
    ->  b.book_id, bks.book_name;
+---------+--------------------+------------------+
| book_id | book_name          | count(b.book_id) |
+---------+--------------------+------------------+
|     201 | Malgudi Days       |                2 |
|     202 | Five Point Someone |                1 |
|     203 | Harry Potter       |                2 |
|     204 | 1984               |                1 |
|     205 | Immortals of Meluha |                2 |
|     206 | Wise and Otherwise |                1 |
|     207 | Da Vinci Code      |                2 |
|     208 | The Blue Umbrella  |                1 |
|     209 | Half Girlfriend    |                1 |
|     210 | Harry Potter 2     |                1 |
+---------+--------------------+------------------+
10 rows in set (0.00 sec)
```

Section 4) mongodb

1)

```
test> use SunbeamDBT
switched to db SunbeamDBT
SunbeamDBT>
```

2)

```
SunbeamDBT> db.books.insertMany([
| {
| id: 2,
| name: "science",
| author: "efg",
| subject: "subject 1",
| price: 200
| },
| {id: 3,
| name: "maths",
| author: "xyz",
| subject: "subject 2",
| price: 300
| },
| {id: 4,
| name: "phy",
| author: "fgq",
| subject: "fjie",
| price: 3232
| },
| {id: 5,
| name: "Exploring C",
| author: "fje",
| }]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6aa62746c3fd89fe32f5fcc7'),
    '1': ObjectId('6aa62746c3fd89fe32f5fcc8'),
    '2': ObjectId('6aa62746c3fd89fe32f5fcc9'),
    '3': ObjectId('6aa62746c3fd89fe32f5fcca')
  }
}
```

3)

```
SunbeamDBT> db.books.deleteOne({name: "Exploring C"});
{ acknowledged: true, deletedCount: 1 }
SunbeamDBT>
```

4)

```
SunbeamDBT> db.books.find({price: {$lt: 400}});
[
  {
    _id: ObjectId('6aa62688c3fd89fe32f5fcc6'),
    id: 1,
    name: 'cpp',
    author: 'abc',
    subjct: 'programming',
    price: 100.5
  },
  {
    _id: ObjectId('6aa62746c3fd89fe32f5fcc7'),
    id: 2,
    name: 'science',
    author: 'efg',
    subject: 'subject 1',
    price: 200
  },
  {
    _id: ObjectId('6aa62746c3fd89fe32f5fcc8'),
    id: 3,
    name: 'maths',
    author: 'xyz',
    subject: 'subject 2',
    price: 300
  }
]
SunbeamDBT>
```

5)

```
SunbeamDBT> db.books.find({price : {$gt: 300}});
[
  {
    _id: ObjectId('6aa62746c3fd89fe32f5fcc9'),
    id: 4,
    name: 'phy',
    author: 'fgq',
    subject: 'fjie',
    price: 3232
  }
]
SunbeamDBT>
```